Questions

Q1.

Which colour of visible light has the longest wavelength?

(1)

☐ **A** blue

☐ **B** green

☐ **C** red

☐ **D** yellow

(Total for question = 1 mark)

Q2.

Answer the question with a cross in the box you think is correct ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Figure 3 shows a ray of light striking a mirror.

The angle of incidence is 30° and the ray of light is reflected.

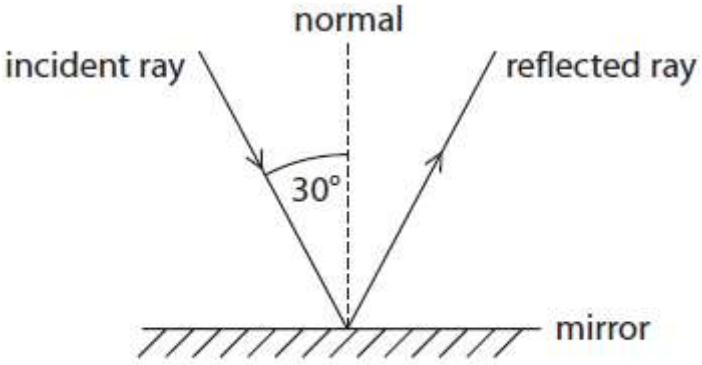


Figure 3

The angle of reflection is the angle between

(1)

☐ **A** the mirror and the incident ray
☐ **B** the mirror and the normal
☐ **C** the reflected ray and the incident ray
☐ **D** the reflected ray and the normal

(Total for question = 1 mark)

Q3.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

This question is about light.

White light includes all the colours in the visible spectrum.

A beam of white light is the only light that shines on a book.
The book appears green.
A red filter is placed between the source of white light and the book.

What colour does the book appear now?

(1)

☐ **A** black

☐ **B** blue

☐ **C** green

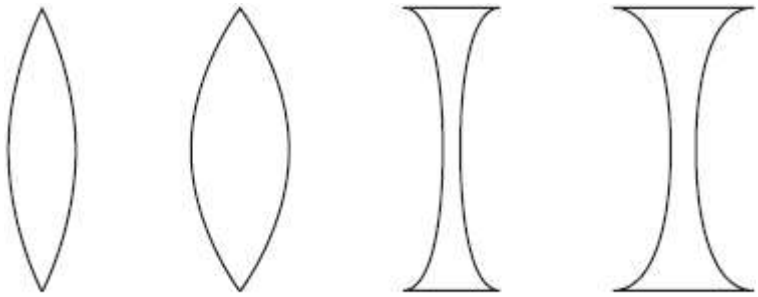
☐ **D** red

(Total for question = 1 mark)

Q4.

(i) Which lens is a converging lens with the greatest power?

(1)



☐ **A** ☐ **B** ☐ **C** ☐ **D**

(ii) The equation that relates the power of a lens to the focal length of the lens is

$$\text{power (in dioptres)} = \frac{1}{\text{focal length (in metres)}}$$

The power of a lens is 5 dioptres.

Use the equation to calculate the focal length of the lens in cm.

(2)

focal length = ... cm

(Total for question = 3 marks)

Q5.

What do all waves in the electromagnetic spectrum have in common?

(1)

☐ **A** the same frequency in a vacuum

☐ **B** the same speed in a vacuum

☐ **C** the same colour in a vacuum

☐ **D** the same amplitude in a vacuum

(Total for question = 1 mark)

Q6.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Different colours of light have different frequencies.

Which of these colours has the highest frequency?

(1)

☐ **A** blue
☐ **B** green
☐ **C** red
☐ **D** yellow

(Total for question = 1 mark)

Q7.

Answer the question with a cross in the box you think is correct (☒). If you change your mind about an answer, put a line through the box (☒) and then mark your new answer with a cross (☒).

Which of these is the best reason for explaining why a leaf appears green in daylight?

(1)

☐ **A** the leaf absorbs green light only
☐ **B** the leaf absorbs yellow and blue light only
☐ **C** the leaf reflects green light only
☐ **D** the leaf reflects yellow and blue light only

(Total for question = 1 mark)

Q8.

All objects emit electromagnetic radiation.

The intensity and wavelength of the emitted radiation vary with the temperature of the object.

Figure 8 shows this variation for an object at two different temperatures.

The visible region of the electromagnetic spectrum is also shown.

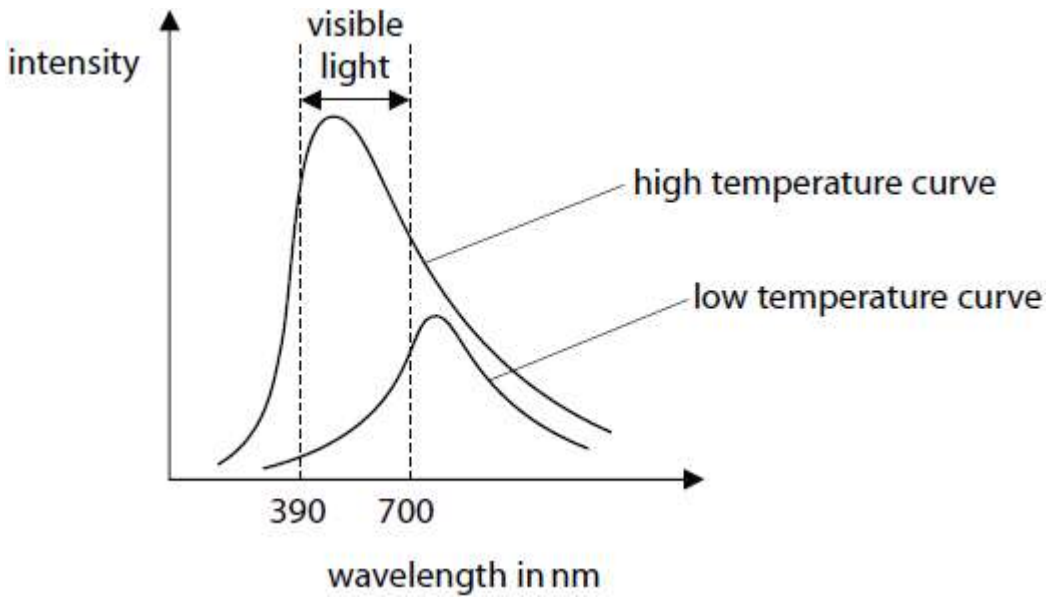


Figure 8

In which part of the electromagnetic spectrum is the peak of the low temperature curve?

(1)

☐ **A** gamma

☐ **B** infrared

☐ **C** radio

☐ **D** ultra violet

(Total for question = 1 mark)

Q9.

Some students repeat Herschel's experiment.
They place four identical thermometers, **P**, **Q**, **R** and **S**, in the shade.
The table shows some of the readings on the thermometers in the shade.
Complete the table to show the reading on thermometer **Q**.

(1)

	thermometers in the shade			
	P	**Q**	**R**	**S**
temperature / °C	18		18	18

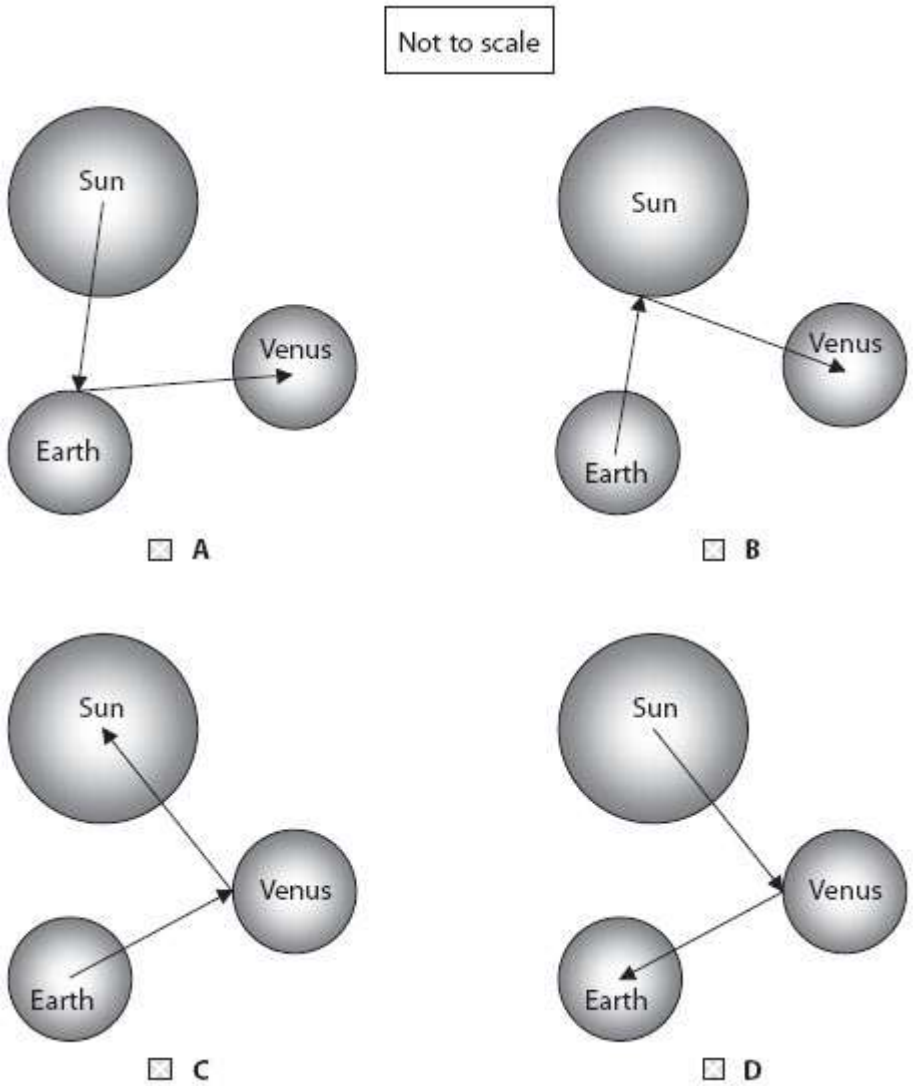
Q10.

Galileo was one of the first scientists to use a telescope to study Venus.

(i) Which of these diagrams best shows how light waves enable us to see Venus?

Put a cross (☒) in the box next to your answer.

(1)



(ii) Use words from the box to complete the sentences.

(2)

asteroids	comets	geocentric	heliocentric
	moons	particle	stars

Galileo also used his telescope to observe the ..... of Jupiter.
His observations provided evidence to support the ..... model of the Solar System.

(iii) Describe how a reflecting telescope is different from the simple telescope which Galileo used.

(2)

.....

.....

.....

.....

Mark Scheme

Q1.

Question Number	Answer	Mark
	C red **The only correct answer is C red** **A** is not correct because blue has a shorter wavelength than red **B** is not correct because green has a shorter wavelength than red **D** is not correct because yellow has a shorter wavelength than red	**(1)**

Q2.

Question Number	Answer	Mark
	D the reflected ray and the normal **A, B** and **C** do not describe the angle of reflection	**(1)** **AO1**

Q3.

Question number	Answer	Mark
	A black B is incorrect as no blue light shines on the object C is incorrect as no green light shines on the object D is incorrect as no red light is reflected from the object	(1) AO3

Q4.

Question Number	Answer	Mark
(i)	 ☐ B **The only correct answer is B** **A** is not correct because it has a smaller power than B **C** is not correct because it is a diverging lens **D** is not correct because it is a diverging lens	(1)

Question Number	Answer	Additional guidance	Mark
(ii)	rearrangement and substitution (1) $\frac{1}{5}$ unit conversion and evaluation (1) 20 (cm)	 award full marks for the correct answer without working accept 0.2 for one mark only	(2)

Q5.

Question number	Answer	Mark
	B	(1)

Q6.

Q7.

Q8.

Question number	Answer	Mark
	B	(1)

Q9.

	Answer	Acceptable answers	Mark
	18 (°C) (1) Ignore any unit given by candidate.		**(1)**

Q10.

	Answer	Acceptable answers	Mark
(i)	D		**(1)**
(ii)	moons (1) heliocentric (1)	must be in correct order	**(2)**
(iii)	A description including two of the following points Reflecting telescope has mirror(s) (1) Galilean telescope has only lenses (1) Reflecting telescope can gather more light / can have a larger objective (1) Image viewed from the side of reflecting telescope (1) Image viewed from end of Galilean telescope. (1)	refracting telescope reverse argument	**(2)**